

# Week 1 - Mockito exercises

## Exercise 1: Mocking and Stubbing

Scenario:

You need to test a service that depends on an external API. Use Mockito to mock the external API and stub its methods.

Steps:

1. Create a mock object for the external API.
2. Stub the methods to return predefined values.
3. Write a test case that uses the mock object.

Solution Code: import static

org.mockito.Mockito.\*; import

org.junit.jupiter.api.Test;

import org.mockito.Mockito;

```
public class MyServiceTest {
```

```
    @Test    public void
```

```
testExternalApi() {
```

```
    ExternalApi mockApi = Mockito.mock(ExternalApi.class);
```

```
    when(mockApi.getData()).thenReturn("Mock Data");
```

```
    MyService service = new MyService(mockApi);
```

```
    String    result    =    service.fetchData();
```

```
    assertEquals("Mock Data", result);
```

```
    }
```

```
}
```

ExternalApi.java package

com; public interface

ExternalApi {

```
String getData();  
}
```

MyService.java package  
com;

```
public class MyService {  
    private ExternalApi api;  
  
    public MyService(ExternalApi api) {  
        this.api = api;  
    }  
  
    public String fetchData() {    return  
api.getData(); // real call will be mocked  
    }  
}
```

MyServiceTest.java package  
com;

```
import static org.junit.jupiter.api.Assertions.assertEquals;  
import org.junit.jupiter.api.Test; import static  
org.mockito.Mockito.mock; import static  
org.mockito.Mockito.when;
```

```
public class MyServiceTest {  
  
    @Test    public void  
testExternalApi() {    // Step  
1: Create a mock
```

```
ExternalApi mockApi = mock(ExternalApi.class);
```

```
// Step 2: Stub method
```

```
when(mockApi.getData()).thenReturn("Mock Data");
```

```
// Step 3: Inject mock into service
```

```
MyService service = new MyService(mockApi);
```

```
// Step 4: Assert expected behavior
```

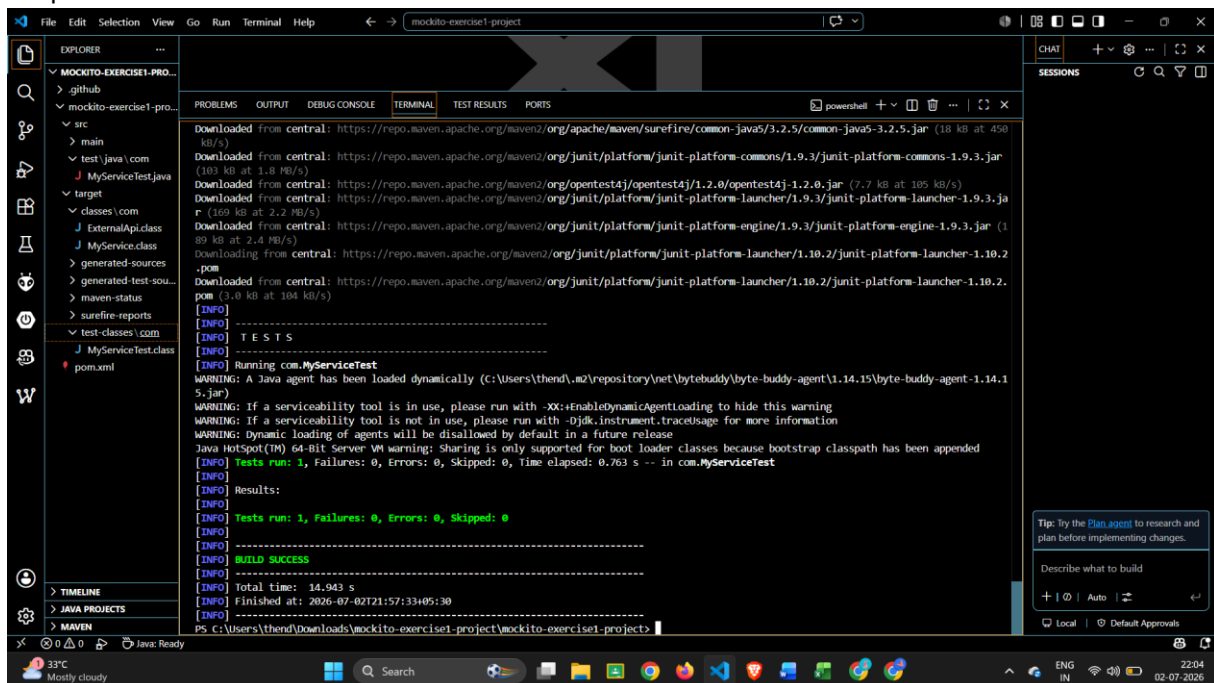
```
String result = service.fetchData();
```

```
assertEquals("Mock Data", result);
```

```
}
```

```
}
```

Output:



## Exercise 2: Verifying Interactions

Scenario:

You need to ensure that a method is called with specific arguments.

Steps:

1. Create a mock object.
2. Call the method with specific arguments.
3. Verify the interaction.

Solution Code:

```
import static org.mockito.Mockito.*; import
org.junit.jupiter.api.Test;
import org.mockito.Mockito;
```

```
public class MyServiceTest {
    @Test    public void
testVerifyInteraction() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        MyService service = new MyService(mockApi);
        service.fetchData();
        verify(mockApi).getData();
    }
}
```

ExternalApi.java

```
public interface ExternalApi {
    String getData();
}
```

MyService.java

```

public class MyService {
    private ExternalApi api;

    public MyService(ExternalApi api) {
        this.api = api;
    }

    public String fetchData() {
        return api.getData();
    }
}

```

MyServiceTest.java

```

public class MyServiceTest {

    @Test    public void
    testVerifyInteraction() {

        // Step 1: Create mock
        ExternalApi mockApi = mock(ExternalApi.class);

        // Step 2: Use the mock in the service    MyService
        service = new MyService(mockApi);    service.fetchData(); //
        This should call mockApi.getData()

        // Step 3: Verify interaction    verify(mockApi).getData(); // This
        passes only if fetchData() was called

    }
}

```

Output:

File Edit Selection View Go Run Terminal Help modkito-exercise2-project

EXPLORER

- MOCKITO-EXERCISE2-PRO...
- > .github
- mockito-exercise2-pro...
- > src
- > main\java\com
- > ExternalApi.java
- > MyService.java
- > test\java\com
- > MyServiceTest.java
- > target
- > pom.xml

MyServiceTest.java

mockito-exercise2-project > src > test > java > com > J MyServiceTest.java > {} com

hackape.com

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL TEST RESULTS PORTS

powerShell + - | | |

```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\thend\Downloads\mockito-exercise2-project\mockito-exercise2-project\src\test\resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ mockito-exercise2-project ---
[INFO] Recompiling the module because of changed dependency.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac (debug target 17) to target\test-classes
[WARNING] location of system modules is not set in conjunction with -source 17
not setting the location of system modules may lead to class files that cannot run on JDK 17
--release 17 is recommended instead of -source 17 -target 17 because it sets the location of system modules automatically
[INFO] --- surefire:3.2.5:test (default-test) @ mockito-exercise2-project ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running com.MyServiceTest
WARNING: A Java agent has been loaded dynamically (C:\Users\thend\.m2\repository\net\bytebuddy\byte-buddy-agent\1.14.15\byte-buddy-agent-1.14.15.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.880 s -- in com.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.837 s
[INFO] Finished at: 2026-07-02T22:13:14+05:30
[INFO]
PS C:\Users\thend\Downloads\mockito-exercise2-project\mockito-exercise2-project>
```

Tip: Try the [Plan agent](#) to research and plan before implementing changes.

+ J MyServiceTest.java

Describe what to build

+ | | Auto | |

Local | Default Approvals

Ln 1, Col 1 Spaces: 2 UTF-8 LF {} Java

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